Department of Artificial Intelligence & Data Science

SmartClass AI: A CNN–LSTM-Based Intelligent Classroom Analytics System

Submitted in partial fulfilment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

In

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

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SRM TRP ENGINEERING COLLEGE

Academic Year: 2026-27 (ODD)

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CERTIFICATE

This is to certify that **Humairah Shiffana A, Anuvarshini B and Harini K** from **Third year AI&DS** have successfully completed their work titled **"SmartClass AI: A CNN–LSTM-Based Intelligent Classroom Analytics System"** at SRM TRP Engineering, Trichy in the partial fulfillment of the Bachelor's of Technology in Artificial Intelligence & Data Science.

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ACKNOWLEDGEMENT

It is with sincere gratitude and deep appreciation that we take this opportunity to acknowledge the invaluable support and guidance we have received throughout the course of this project. This journey has been both intellectually enriching and personally rewarding, significantly enhancing our understanding of deep learning and its practical applications in education. We would like to express our heartfelt thanks to our guide, Mrs. R. Kalaiyarasi, and to our Head of the Department, Dr. R. Perumal Raja, for their guidance, encouragement, and constructive feedback during the development of this project. Their insights and support were instrumental in shaping the direction and quality of our work. We also extend our appreciation to the department for fostering an environment that promotes learning, innovation, and academic growth. The resources and opportunities provided played a vital role in the successful completion of this project. A special note of thanks goes to the supporting staff for their cooperation and timely assistance, which ensured that the process remained smooth and efficient at every stage. Finally, we would like to express our deepest gratitude to our parents for their constant support, encouragement, and belief in our abilities. Their motivation has been a source of strength throughout this journey. This project has not only strengthened our knowledge but has also emphasized the importance of dedication, perseverance, and continuous learning. We remain sincerely thankful to everyone who contributed, directly or indirectly, to the successful completion of this project.

Academic Year: 2026-2027

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ABSTRACT

Understanding how students behave and respond during a lecture is a persistent challenge in modern classrooms, where a single instructor cannot continuously observe every learner. This project presents SmartClass AI, an intelligent classroom analytics system that uses deep learning to automatically recognise students, estimate their emotional state, and measure their engagement level from classroom video. The proposed system combines a Convolutional Neural Network (CNN) with a Long Short-Term Memory (LSTM) network. The CNN extracts spatial facial features from every detected face frame, while the LSTM models how these features change across a short sequence of frames, allowing the system to understand behaviour over time rather than from a single image. Video frames are captured from a classroom camera, faces are detected and cropped using OpenCV, and each face is matched against enrolled students so that attendance is marked automatically. The cropped faces are resized, normalised and grouped into fixed-length sequences before being passed to the CNN–LSTM model, which outputs an emotion class and an engagement level for each student. The model is trained on labelled emotion and engagement data such as FER-2013 and DAiSEE, using categorical cross-entropy loss and the Adam optimiser, and is evaluated using accuracy, precision, recall, F1-score and confusion matrices. Individual predictions are then aggregated into class-level statistics such as attendance, engagement over time, emotion distribution and attention by seating zone, and are displayed on an interactive dashboard so that faculty can identify disengaged learners and difficult portions of a lecture and adapt their teaching accordingly. The project shows that a CNN–LSTM pipeline is a practical and scalable way to turn raw classroom video into meaningful teaching insights, while factors such as lighting, camera angle, occlusion and student privacy must be handled carefully.

INTRODUCTION

Introduction: SmartClass AI – CNN–LSTM Based Classroom Analytics

Education is increasingly supported by digital technology, yet the most valuable signal in a classroom, namely how students actually feel and respond while a topic is being taught, still depends almost entirely on the teacher's own observation. In a class of thirty or more students it is impossible for an instructor to notice every confused, bored or distracted learner while also teaching. Attendance is usually taken manually, which consumes lecture time, and feedback about the quality of a lecture normally arrives only after the exam results are known. An automated system that can observe the classroom continuously and summarise what it sees can therefore give teachers timely and objective insight.

Computer vision and deep learning have made such a system practical. Convolutional Neural Networks (CNNs) learn hierarchical visual features directly from images, from low-level edges and textures to high-level facial structures, and are the standard tool for face detection, face recognition and facial expression analysis. However, a single image is a poor basis for judging engagement. A student who looks down for one frame may be taking notes, while a student who stares at the board for a long time may have stopped following the lesson. Engagement is a property of behaviour over time. Long Short-Term Memory (LSTM) networks are a type of recurrent neural network with gated memory cells that learn long-range dependencies in sequences, which makes them well suited to modelling how facial features evolve from frame to frame.

SmartClass AI combines these two ideas in a single CNN–LSTM architecture. A CNN encoder processes every face frame in a short clip and produces a compact feature vector. The sequence of vectors is passed to an LSTM that captures temporal patterns, and two output layers then predict the student's emotion and engagement level. The overall problem is divided into four tasks: face detection and recognition for automatic attendance, emotion recognition, engagement estimation, and classroom-level analytics presented on a dashboard.

The effectiveness of such a system depends heavily on the quality and diversity of its training data. Public datasets such as FER-2013, which contains labelled facial expression images, and DAiSEE, which contains video clips of learners annotated with affective states including engagement, provide a wide range of faces, lighting conditions and expressions. Preprocessing steps such as face detection, cropping, resizing, normalisation and data augmentation are applied to make the model robust to the conditions found in a real classroom.

Several challenges must be addressed. Faces may be partly hidden by hair, masks or hands, students seated at the back appear smaller and darker, and expression datasets are often imbalanced. Student privacy is equally important: the system is intended to be used with the knowledge and consent of the students and the institution, and the dashboard reports aggregate class-level insights rather than judging individuals. Designing an effective model therefore requires careful choices of architecture, loss functions and training strategy, which are described in the following chapters.

SYSTEM REQUIREMENTS

Category	Requirement	Description
Operating System	Windows / Linux / macOS	A compatible OS that supports deep learning frameworks. Linux is preferred for better performance and GPU support.
Processor (CPU)	Intel i5/i7 or AMD Ryzen 5/7 (or higher)	Multi-core processors help in faster data preprocessing and model execution.
GPU (Recommended)	NVIDIA GPU (e.g., GTX 1650 / RTX 2060 / RTX 3060 or higher)	Required for accelerating CNN–LSTM training using CUDA. Significantly reduces training time compared to CPU.
RAM	Minimum 8 GB (Recommended: 16 GB or more)	Required to handle large image datasets and batch processing during training.
Storage	Minimum 20–50 GB free space	Needed for storing datasets (images and video clips), trained models, and project files.
Camera	HD webcam / classroom IP camera (720p or higher)	Captures classroom video frames used for face detection, recognition, and emotion and engagement analysis.
Programming Language	Python	Widely used for deep learning due to simplicity and extensive library support.
Deep Learning Framework	TensorFlow / Keras	Used to build, train, and evaluate the CNN–LSTM model for emotion and engagement prediction.
Computer Vision Library	OpenCV, face_recognition (dlib)	Used for frame extraction, face detection, cropping, resizing, and identifying students for attendance.
Data Handling Libraries	NumPy, Pandas	Used for efficient data manipulation and preprocessing.
Visualization & Dashboard	Matplotlib, Plotly, Streamlit	Used for plotting training curves and for building the interactive classroom analytics dashboard.
Development Environment	Visual Studio Code / Jupyter Notebook	Used for writing, testing, and debugging code.
Dataset	FER-2013 / DAiSEE + classroom images	Labelled facial expression images and engagement video clips used for training and evaluation, plus enrolled student face images.

METHODOLOGY

Methodology: SmartClass AI using CNN–LSTM

The methodology of this project focuses on designing and implementing a deep learning-based system that analyses classroom video and reports attendance, emotion and engagement. The approach follows a structured pipeline that includes video capture, face detection and recognition, preprocessing, model design, training, evaluation, analytics and deployment. A Convolutional Neural Network (CNN) learns the spatial features of each face, and a Long Short-Term Memory (LSTM) network learns how those features change over time, so that engagement is judged from behaviour across a clip rather than from a single frame. The methodology ensures that the model is trained efficiently and produces meaningful predictions in real classroom conditions.

1. Overall System Workflow

The workflow begins with capturing classroom video and detecting faces in each frame. Detected faces are recognised against the enrolled students to mark attendance, and the cropped faces are grouped into short sequences. The CNN–LSTM model predicts emotion and engagement for each student, and an analytics module aggregates the results into class-level statistics that are shown on a dashboard.

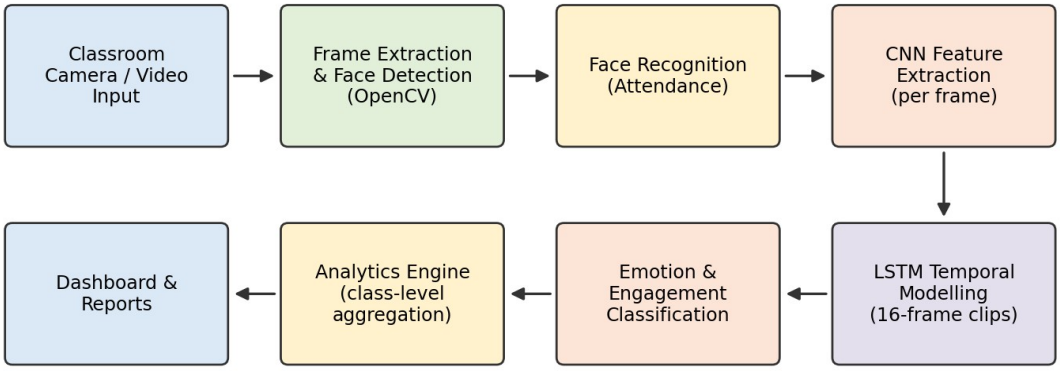


Figure 1: Overall workflow of the SmartClass AI system

2. Data Collection and Dataset Selection

The first step involves selecting suitable datasets. FER-2013 contains tens of thousands of small grayscale face images labelled with seven basic emotions (angry, disgust, fear, happy, sad, surprise and neutral). DAiSEE (Dataset for Affective States in E-learning Environments) contains over nine thousand video clips of learners annotated with the states boredom, confusion, engagement and frustration, each at four intensity levels. Together they supply the emotion labels and the temporal engagement labels needed by the model. Face images of the enrolled students are collected separately for the attendance module.

Key dataset characteristics:

- Large number of labelled face images and video clips
- Diversity in expressions, lighting, pose, and background

- Frame sequences that capture behaviour over time
- Real-world conditions for better generalisation

A well-balanced dataset is essential to avoid bias and improve model performance.

3. Data Preprocessing

Before feeding the data into the model, preprocessing is performed to ensure consistency. Since deep learning models are sensitive to the input format, all frames must be standardised.

Key preprocessing steps include:

- **Face Detection:** Using OpenCV to detect and crop face regions from every frame
- **Resizing and Grayscale Conversion:** Converting crops to a fixed size of 64×64 pixels
- **Normalisation:** Scaling pixel values between 0 and 1
- **Sequence Building:** Grouping 16 consecutive frames of a student into one clip
- **Data Augmentation:** Applying rotation, flipping, brightness change and zooming

These steps improve model robustness and help prevent overfitting.

4. Face Detection and Recognition

Each frame from the classroom camera is scanned for faces using OpenCV. For attendance, every detected face is converted into a numerical face embedding and compared with the embeddings stored for the enrolled students. If the distance to the closest enrolled student is below a set threshold, the student is marked present; otherwise the face is labelled as unknown. A student is marked present only once per session, and the same identity is used to link the face sequences that are sent to the emotion and engagement model.

5. CNN–LSTM Model Architecture

The core of the system is a CNN–LSTM network. A CNN encoder made of three convolution and pooling blocks is applied to every frame of the clip using a TimeDistributed layer, so the same weights are shared across all frames. Each frame is reduced to a 128-dimensional feature vector, and the sequence of 16 vectors is passed to an LSTM layer with 128 units that captures temporal patterns such as sustained attention, drifting focus or a change of expression. Dropout layers reduce overfitting. The model uses a **multi-task learning approach**, where shared layers feed two separate output layers.

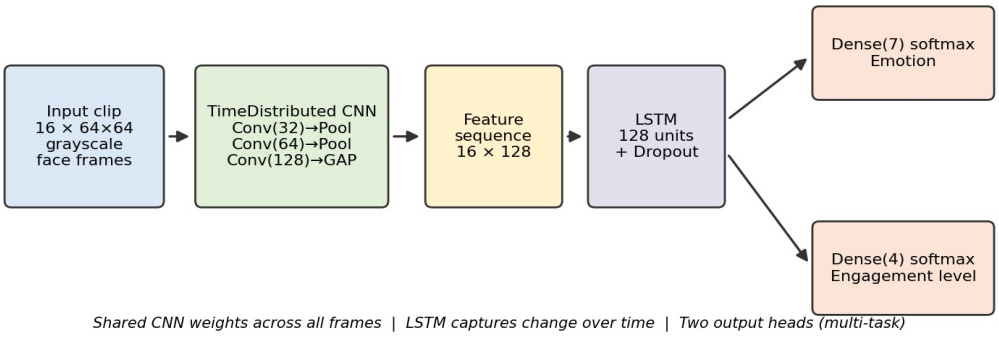


Figure 2: CNN–LSTM architecture used in SmartClass AI

Model components:

- **Convolutional Layers:** Extract spatial features such as edges, textures and facial patterns
- **Pooling Layers:** Reduce dimensionality and computational cost
- **LSTM Layer:** Learns how features change across the frames of a clip
- **Fully Connected Layers:** Produce the final predictions

Output layers:

- Emotion → Seven-class classification (softmax)
- Engagement → Four-level classification (softmax)

This design allows the model to learn both tasks simultaneously from the same shared features.

6. Model Training

The model is trained using the preprocessed data, where it learns to map each face sequence to its emotion and engagement labels. The data is divided into training, validation, and testing sets, and the validation set is used to monitor overfitting during training.

Key training components:

- **Loss Functions:** Categorical cross-entropy for emotion and for engagement level
- **Optimizer:** Adam optimizer for efficient gradient descent
- **Batch Size & Epochs:** Tuned based on dataset size and validation performance
- **Regularization:** Dropout layers and early stopping to prevent overfitting
- **Class Weights:** Used to reduce the effect of imbalanced emotion classes

During training, the model updates its weights through backpropagation through time to minimise prediction error.

7. Model Evaluation

After training, the model is evaluated on the unseen test set to measure its performance. Both outputs are classification tasks, so the same family of metrics is used for each:

- **Emotion Prediction:** Accuracy, Precision, Recall, F1-score
- **Engagement Prediction:** Accuracy, macro F1-score, and per-level recall

Visualisation tools such as confusion matrices and training and validation curves are used to analyse performance and identify possible improvements, for example classes that are frequently confused with each other.

8. Classroom Analytics and Dashboard

Per-student predictions are aggregated by an analytics module into class-level information. The dashboard presents the number of students present and the attendance percentage, the average engagement and its variation through the lecture, the distribution of detected emotions, the share of students at each engagement level, and an attention heatmap by seating zone. Alerts are raised when the average engagement falls below a threshold, helping the teacher decide when to pause, revisit a concept or change the activity.

9. Prediction and Deployment

Once the model is trained and evaluated, it can be used for real-time prediction on a live camera feed or a recorded lecture. The system performs the following steps:

- Capture of video frames from the classroom camera
- Face detection, recognition and preprocessing
- Model inference on 16-frame sequences
- Display of attendance, emotion and engagement on the dashboard

The trained model can be deployed on a classroom computer or a local server so that video does not need to leave the institution.

10. Challenges and Optimization

Some challenges encountered in this kind of system include:

- Variation in lighting, camera angle and distance from the camera
- Partial occlusion of faces and small faces at the back of the class
- Imbalanced emotion classes and subjective engagement labels
- Student privacy and the need for consent

To overcome these challenges, techniques such as data augmentation, transfer learning, class weighting, hyperparameter tuning, and storing only face embeddings and aggregate statistics rather than raw video can be applied.

11. Summary of Methodology

In summary, the methodology involves a complete pipeline starting from video capture and preprocessing to face recognition, CNN–LSTM training, evaluation, analytics and dashboard deployment. By using a CNN to learn spatial facial features and an LSTM to learn temporal behaviour, the system turns classroom video into attendance, emotion and engagement information that supports better teaching decisions.

IMPLEMENTATION

Implementation of SmartClass AI

SmartClass AI is implemented in Python using TensorFlow/Keras for the deep learning model, OpenCV for video processing, and Streamlit with Matplotlib and Plotly for the dashboard. The project is organised into small modules, each responsible for one stage of the pipeline.

Project modules:

- **face_utils.py:** Reads video frames, detects faces with OpenCV, and crops and normalises them to 64×64 grayscale images
- **attendance.py:** Generates face embeddings, matches them with the enrolled students, and records attendance
- **smartclass_model.py:** Defines the CNN–LSTM network with its emotion and engagement output layers
- **train.py:** Loads the datasets, builds 16-frame sequences, trains the model and saves the best weights
- **app.py:** Runs the live pipeline and shows the results on the analytics dashboard

Execution steps:

- The classroom camera or a recorded lecture is opened and frames are read continuously
- Faces are detected, recognised, and collected into sequences of 16 frames per student
- Each sequence is passed to the CNN–LSTM model to predict emotion and engagement
- Predictions are aggregated for the whole class and displayed on the dashboard

The source code of the model is shown in Annexure A, the classroom analytics dashboard is shown in Annexure B, and the face recognition output is shown in Annexure C. The dashboard and face recognition views present the output expected from the system.

CONCLUSION

The implementation of SmartClass AI demonstrates how deep learning can turn raw classroom video into useful teaching insight. By combining a Convolutional Neural Network, which learns spatial facial features from each frame, with a Long Short-Term Memory network, which learns how those features change over time, the system is able to recognise students for automatic attendance and to estimate emotion and engagement from short sequences of frames rather than from isolated images. A structured methodology was followed, beginning with dataset selection and preprocessing, followed by model design, training, evaluation and the construction of a classroom dashboard. The multi-task design, in which emotion and engagement are predicted from shared features, proved to be an efficient way to obtain several useful outputs from a single model. The dashboard presents attendance, engagement over time, emotion distribution and attention by seating zone, so that a teacher can quickly see when the class is losing focus and respond.

In the future, the system can be extended with pose and gaze estimation, audio cues and multiple camera views, trained on data collected from the institution's own classrooms, and integrated with the college's attendance and learning-management systems. Throughout, the system should be used responsibly, with the consent of students and with a focus on class-level insights that support teaching rather than on monitoring individuals.

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- Streamlit Documentation – https://docs.streamlit.io
- Deep Learning Book – Ian Goodfellow, Yoshua Bengio, Aaron Courville – https://www.deeplearningbook.org

ANNEXURE A – SOURCE CODE

The screenshot below shows the implementation of the SmartClass AI model in Visual Studio Code. The TimeDistributed CNN encoder extracts features from each frame, the LSTM layer models the sequence, and two softmax output layers predict emotion and engagement.

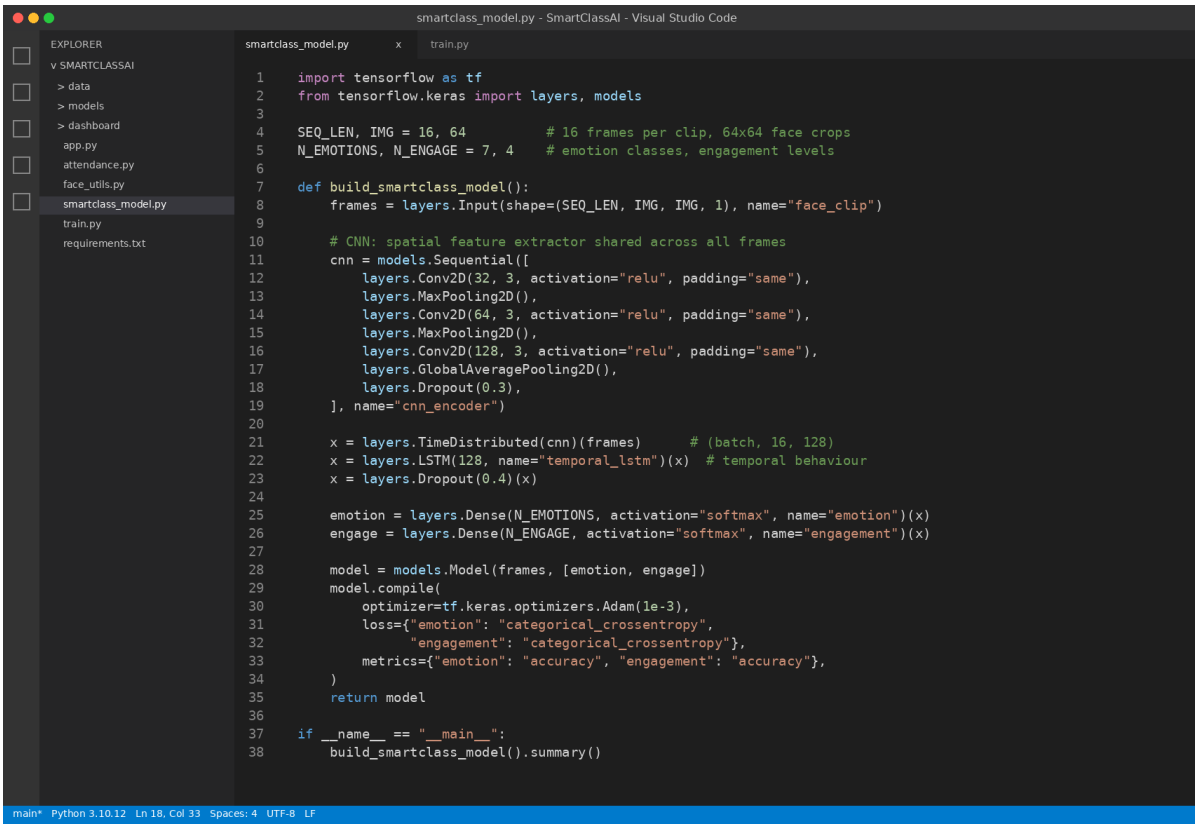
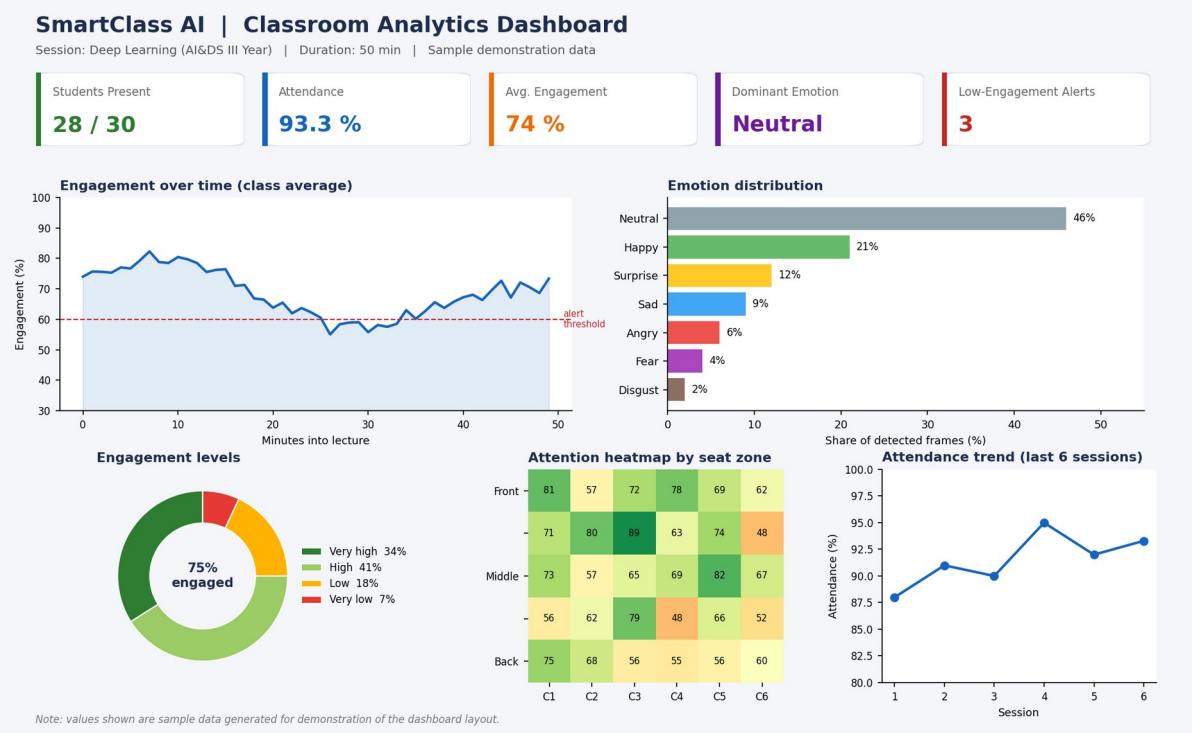


Figure A: Source code of the CNN–LSTM model (smartclass_model.py) in Visual Studio Code

ANNEXURE B – CLASSROOM ANALYTICS DASHBOARD

The dashboard summarises the generalised analysis of a classroom session: attendance, average engagement, emotion distribution, engagement levels, an attention heatmap by seating zone, and the attendance trend across sessions.



ANNEXURE C – FACE RECOGNITION OUTPUT

The figure shows a sample of the face recognition stage running on a laptop webcam in Visual Studio Code. The detected face is enclosed in a bounding box and labelled with the recognised person's name. SmartClass AI uses the same detection and recognition approach to mark attendance and to pass each student's face sequence to the CNN–LSTM model.

